

⇒ Heat Capacity:

Heat energy required to raise 1°C rise in temperature of given amount of a substance is called heat capacity.
(temperature dependent)

⇒ Specific heat capacity:

Heat energy required to raise 1°C rise in temperature of 1g of a substance is called specific heat capacity.

⇒ Molar heat capacity:

Heat energy required to raise 1°C rise in temperature of 1mol of a substance is called molar heat capacity.

Heat Capacity:

Mathematically

$$C = \frac{dq}{dT} \rightarrow (i)$$

$$C = \frac{dE + PdV}{dT} \quad (\because dq = dE + PdV)$$

$$C = \frac{dE}{dT} + \frac{PdV}{dT} \rightarrow (ii)$$

⇒ Heat Capacity at constant volume:

From eq. (ii)

$$C_v = \left(\frac{dE}{dT} \right)_v + 0 \quad (\because dv = 0)$$

$$C_v = \left(\frac{dE}{dT} \right)_v$$

small change
to convert it
to large
change we'll
integrate

$$\leftarrow dE = C_v dT$$

Integrating both sides

$$\int_{E_1}^{E_2} dE = \int_{T_1}^{T_2} C_v dT$$

$$[E]_{E_1}^{E_2} = C_v [T]_{T_1}^{T_2}$$

$$E_2 - E_1 = C_v (T_2 - T_1)$$

$$\Rightarrow \boxed{\Delta E = C_v \Delta T}$$

⇒ Heat Capacity at constant Pressure:

$$\Rightarrow C_p = \left(\frac{dE}{dT} \right)_p + P \left(\frac{dv}{dT} \right)_p \rightarrow (i)$$

$$\text{As } H = E + PV$$

Differentiating b.s w.r.t T at const P

$$\left(\frac{dH}{dT} \right)_p = \left(\frac{dE}{dT} \right)_p + P \left(\frac{dv}{dT} \right)_p \rightarrow (ii)$$

Comparing eq. (i) & (ii)

$$C_p = \left(\frac{dH}{dT} \right)_p \Rightarrow dH = C_p dT$$

$$\int_{H_1}^{H_2} dH = \int_{T_1}^{T_2} C_p dT \Rightarrow [H]_{H_1}^{H_2} = C_p [T]_{T_1}^{T_2}$$

$$\boxed{\Delta H = C_p \Delta T}$$

⇒

Electromagnetic spectrum:

The arrangement of radiations of different electromagnetic waves on the basis of their frequency is called electromagnetic spectrum.

⇒

cosmic rays > γ -rays > x-rays > UV rays
> visible > infrared > microwaves > radiowaves

↓

visible rays

range

$3600\text{\AA} - 7600\text{\AA}$

⇒ Types of spectrum:

i) Continuous spectrum:

In electromagnetic radiations, if different colors are diffused or mixed to each other then it is called continuous spectrum.

— sunlight and prism example

ii) Discontinuous spectrum:

The type of spectrum in which different colors are separated from each other is called discontinuous spectrum.

iii) Emission spectrum:

The emission of light from an object when its atoms are excited, resulting in the release of energy in the form of light is called emission spectrum.

— on heating electrons jump from lower orbit to higher orbit by gaining energy (excited). Then they go back to their original pos. by releasing energy in form of light

iv) Absorption spectrum:

When light passes through a solution, some part of it is absorbed by the solution and some is passed which goes to screen. The spectrum obtained on screen is called absorption spectrum.

v) Atomic / line spectrum:

The spectrum obtained by interaction of electromagnetic radiations with atoms is called atomic spectrum.

— The emitted light by interaction will give spectrum and is shown as bright light.

vi) Molecular / band spectrum:

The spectrum obtained by interaction of electromagnetic radiations with molecule is called molecular spectrum.

— Some energy changes vibrational as well as rotational motion (also emission of light

— this shows band of lines.

⇒ Types of molecular spectrum:

- rotational spectroscopy / microwave spectroscopy
(when electromagnetic radiations pass)
(some will change rotational motion...)
- vibrational spectroscopy / IR spectroscopy
(.... some change the)
(vibrational motion)
- electronic spectroscopy / UV or visible spectroscopy
(.... some change the)
(electronic energy levels)
- translational spectroscopy
(.... some changes rotational, vibrational,
(electronic & translational energies))

Energies

trans < rotat < vib < electronic

UV > Visible > IR > Micro > Radio

┌
└
both change the
electronic
rotation

┌
└
changes
rotational

changes
~~rotational~~
vibrational

┌
└
changes
translational

13-01-25

GC-Th

Monday

Adsorption: The phenomenon in which gas or liquid molecule concentrate or accumulate on solid or liquid surface is called adsorption.

— In adsorption, the molecule doesnot penetrate inside the solid or liquid.

Absorption: In absorption, the molecules also penetrate inside the body (of solid/liquid) and as on the surface...

Sorption: The phenomenon in which both adsorption and absorption occur simultaneously i.e molecules are accumulating on surface as well as penetrating inside of it.

⇒ The molecules accumulating are called adsorbate.
↓
on surface of solid or liquid

⇒ The substance whose surface is accumulated by gas or liquid molecules is called absorbent.

e.g charcoal + gases }
charcoal dip into } adsorbance
acetic acid + water }

⇒ Adsorption Isotherm:

The relation which tells how much adsorbate is adsorbed on the surface of adsorbent at a particular temperature is called adsorption isotherm.

⇒ Langmuir Adsorption Isotherm:

- he derived a relationship of adsorption of gases with surface of solid.
- LAI does not tell about liquid adsorption
- In 1916, Langmuir ~~proved~~ proposed the model about how much gas is ~~absorbed~~ adsorbed on how much liq surface.

— Postulates of that model

- ① Solid surface on which adsorption of gases occur is homogeneous.
- ② Adsorption sites on surface of ~~liquid~~ ^{solid} are fixed.
- ③ The energy of each adsorption site at surface of solid is same.
- ④ Adsorption are localized in nature
(gas molecules are localized, they cannot move from one site to another)
- ⑤ Condensation and evaporation of gases take place simultaneously.
OR Both adsorbance and desorbance take place simultaneously.
- ⑥ Adsorption & desorption take place at equilibrium

Mathematical Derivation of L.A.T:

- Initially when gas molecules are in space, the adsorption sites are free. So, desorption can't happen.
- So initially the rate of adsorption is higher than rate of desorption.
- With the passage of time, rate of adsorption decrease and rate of desorption increase and at last rate of adsorption becomes equal to rate of desorption, which is called equilibrium.
- Fraction of occupied sites is represented by θ which is
$$\theta = \frac{\text{occupied sites}}{\text{available sites}}$$

When all sites are covered then fraction will be 1. Then based will be $(1-\theta)$.

$\Rightarrow$

rate of adsorption $\propto$ based surface $\times P$

rate of adsorption $\propto (1-\theta) P$

rate of adsorption $= k_1 (1-\theta) P \rightarrow (i)$

$\Rightarrow$

rate of desorption $\propto \theta$

rate of desorption $= k_2 \theta \rightarrow (ii)$

$\Rightarrow$ At equilibrium (or postulate of LAI)

rate of adsorption = rate of desorption

$$k_1 (1-\theta) P = k_2 \theta$$

$$k_1 P - k_1 \theta P = k_2 \theta$$

$$k_1 P = k_1 \theta P + k_2 \theta$$

$$k_1 P = \theta (k_1 P + k_2)$$

$$\theta = \frac{k_1 P}{k_1 P + k_2}$$

$$\theta = \frac{\frac{k_1 P}{k_2}}{\frac{k_1 P}{k_2} + 1}$$

$$\theta = \frac{\frac{k_1 P}{k_2}}{1 + \frac{k_1 P}{k_2}}$$

$$\boxed{\theta = \frac{bP}{1+bP}} \quad \text{where } b = \frac{k_1}{k_2}$$

$\downarrow$
 (iv)

$\Rightarrow$ Unit mass of adsorbent (solid) = m

Amount of adsorbate (gas molec) = x

$$y = \frac{x}{m}$$

Where y is amount of gas molecules per unit mass of adsorbent.

So,

$$\frac{x}{m} \propto \theta$$

or $y \propto \theta$

$$\Rightarrow \boxed{y = k\theta} \rightarrow (v)$$

Putting value of θ in eq (5) fro (4)

$$y = \frac{k bP}{1+bP}$$

$$\Rightarrow \boxed{y = \frac{aP}{1+bP}} \rightarrow (vi)$$

where $a = kb$

From eq. (iv) $\theta = \frac{bP}{1+bP}$

Case-1 (At V. high Pressure)

When P is very high then bP will also become larger then 1 can be negligible. So,

$$\theta = \frac{bP}{bP} \quad \left(\because 1+bP \approx bP \right)$$

$\therefore bP \gg 1$

$$\boxed{\theta = 1}$$

Case-2 (At V. low Pressure)

When P is very low then bP will become more smaller and can be negligible

i.e. $(1+bP \approx 1 \therefore 1 \gg bP)$

So,

$$\theta = bP$$

$\Rightarrow$

$$\boxed{\theta \propto P}$$

⇒ Test of L.A.I.
As

$$y = \frac{aP}{1+bP}$$

$$\frac{y}{P} = \frac{aP}{1+bP} \bigg/ P$$

$$= \left(\frac{aP}{1} + \frac{aP}{bP} \right) \bigg/ P$$

$$= \left(\frac{aP}{1} + \frac{a}{b} \right) \bigg/ P$$

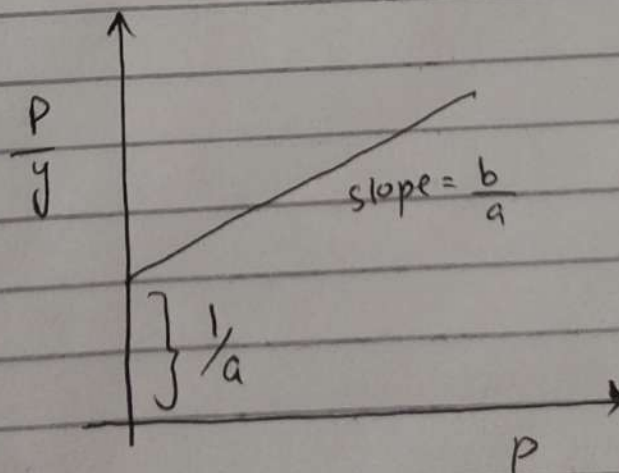
$$\frac{y}{P} = \frac{aP}{P} + \frac{a}{bP}$$

$$\frac{y}{P} = a + \left(\frac{a}{b} \right) \frac{1}{P}$$

⇒

$$\frac{P}{y} = \frac{1}{a} + \left(\frac{b}{a} \right) P$$

$$\begin{array}{ccccccc} \downarrow & & \downarrow & & \downarrow & & \downarrow \\ y & = & c & + & m & x \end{array}$$



← P.L.C. →

06-01-25

G.C - Th

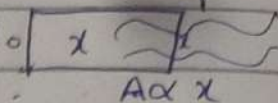
Monday

Beam of parallel radiations

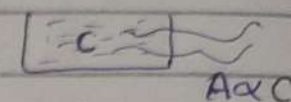
Law of Absorption:

Substances, absorbing material or dilute sol absorb radiations according to some laws that are called laws of absorption

Absorbing material



Dilute solution



When radiations pass through dilute sol or absorbing material some are passed & some are absorbed.

⇒ Lamberts law:

According to this law

"Absorbance of radiation is directly proportional to the thickness of absorbing material"

Monochromatic radiations are

"When beam of parallel monochromatic radiations are passed perpendicularly to homogeneous absorbing material/medium then absorbance is directly proportional to thickness of absorbing material."

$$A \propto x$$

⇒ Beer's law:

"Absorbance of radiation is directly proportional to concentration of dilute solution."

"When beam of parallel monochromatic radiations are passed perpendicularly to the dilute solution, then absorbance is directly proportional to the ~~thickness~~ concentration of dilute sol"

$$A \propto C$$

Deriv $\Rightarrow$ Lambert's Law (proof):

rate of decrease in intensity w.r.t thickness is directly proportional to intensity of radiations passed to the absorbing material.

$$-\frac{dI}{dx} \propto I$$

$$-\frac{dI}{dx} = kI$$

$$-\frac{dI}{I} = k dx$$

Integrating both sides

$$-\int \left(\frac{1}{I}\right) dI = \int k dx$$

As at $x=0$, $I=I_0$ and at $x=x$, $I=I$. So, by applying boundary conditions.

$$-\int_{I_0}^I \left(\frac{1}{I}\right) dI = k \int_0^x 1 dx$$

$$-\ln[I]_{I_0}^I = k[x]_0^x$$

$$-\ln I + \ln I_0 = kx$$

$$\ln I_0 - \ln I = kx$$

$$\ln \frac{I_0}{I} = kx$$

$$2.303 \log \frac{I_0}{I} = kx$$

$$\log \frac{I_0}{I} = \frac{k}{2.303} x \Rightarrow \log \frac{I_0}{I} = \epsilon x$$

(Optical density) $\leftarrow$ or Absorbance $\swarrow$ absorption coefficient / extension coefficient $\searrow$

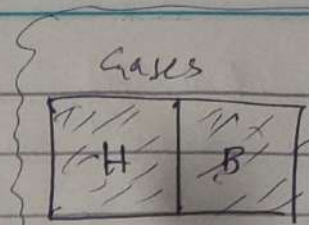
$$A = \epsilon x$$

$$\boxed{A \propto x}$$

proved

⇒ Entropy (S):

Measure of disorderliness of a substance is called entropy.



"The energy of any natural process in universe is conserved while entropy of any natural process in universe ~~is~~ increases."

This the 2nd law of thermodynamics in terms of entropy.

⇒ Physical Significance of entropy:

- i) Mixing of gases
- ii) Dissolution of salt in water
- iii) Conversion of solid to liquid or liquid to gas.

Mathematical formulation of entropy:

"The ratio of heat energy absorbed by substance ~~is~~ and temperature at which heat is absorbed

⇒

$$S = \frac{dq_{rev}}{T}$$

"Ratio of amount of heat absorbed by body in reversible manner at absolute temperature is called entropy."

Unit:

cal. degree⁻¹

cal. degree⁻¹ . mol⁻¹

T K⁻¹

(if amount of substance is 1 mol)

Entropy depends on initial and final state so it is state function.

⇒ Entropy change of ideal gas:

$$dS = \frac{dq_{rev}}{T}$$

⇒

$$dS = \frac{dE + PdV}{T}$$

First law of thermodynamics
 $dq = dE + PdV$

$$dS = \frac{nC_v dT + \frac{nRT}{V} dv}{T} \quad \text{As}$$

$$pV = nRT$$
$$p = \frac{nRT}{V}$$

$$dS = \frac{nC_v dT}{T} + \frac{nRT}{TV} dv$$

$$dS = \frac{nC_v dT}{T} + \frac{nR dv}{V}$$

$$\int_{S_1}^{S_2} dS = \int_{T_1}^{T_2} \frac{nC_v dT}{T} + \int_{V_1}^{V_2} \frac{nR dv}{V}$$

$$\int_{S_1}^{S_2} dS = nC_v \int_{T_1}^{T_2} \frac{1}{T} \cdot dT + nR \int_{V_1}^{V_2} \frac{1}{V} \cdot dV$$

$$[S]_{S_1}^{S_2} = nC_v [\ln T]_{T_1}^{T_2} + nR [\ln V]_{V_1}^{V_2}$$

$$S_2 - S_1 = nC_v [\ln T_2 - \ln T_1] + nR [\ln V_2 - \ln V_1]$$

Ans. rev

~~Equation~~

For n mol of IC₄

$$\Delta S = nC_v \left(\ln \frac{T_2}{T_1} \right) + nR \left(\ln \frac{V_2}{V_1} \right) \rightarrow (i)$$

$$\Delta S = nC_v \left(2.303 \log \frac{T_2}{T_1} \right) + nR \left(2.303 \log \frac{V_2}{V_1} \right)$$

For 1 mol of IC₄

$$\Delta S = C_v \left(\ln \frac{T_2}{T_1} \right) + R \left(\ln \frac{V_2}{V_1} \right) \rightarrow (ii)$$

In terms of Pressure (P)

For n mol of Ideal Gas

$$\Delta S = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{P_1 T_2}{P_2 T_1}$$

$\Rightarrow$

$$\begin{aligned} \Delta S & \\ \frac{P_1 V_1}{T_1} &= \frac{P_2 V_2}{T_2} \\ \frac{P_1 T_2}{P_2 T_1} &= \frac{V_2}{V_1} \end{aligned}$$

$$\Delta S = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$$

$$= \ln \frac{T_2}{T_1} (nC_v + nR) + nR \ln \frac{P_1}{P_2}$$

$$\Delta S = n \ln \frac{T_2}{T_1} (C_v + R) + nR \ln \frac{P_1}{P_2}$$

$\Rightarrow$

$$\Delta S = n \ln \frac{T_2}{T_1} C_p + nR \ln \frac{P_1}{P_2} \rightarrow (iii)$$

$$C_p - C_v = R$$

$$C_v + R = C_p$$

Case - I (Isochoric)

$$\Delta S = nC_v \ln \ln 1 \quad \text{At } V_2 = V_1 = V$$

$$\Delta S = nC_v \ln \frac{T_2}{T_1} + nR \ln 1$$

$$\boxed{\Delta S = nC_v \ln \frac{T_2}{T_1}} \quad (\because \ln 1 = 0)$$

Case - II (Isobaric)

$$\boxed{\Delta S = nC_p \ln \frac{T_2}{T_1}}$$

Case - III (Isothermal)

$$\Delta S = nC_v \ln 1 + nR \ln \frac{V_2}{V_1}$$

$$\boxed{\Delta S = nR \ln \frac{V_2}{V_1}} \quad (\because \ln 1 = 0)$$

and eq. (ii) becomes

$$\Delta S = nC_p \ln 1 + nR \ln \frac{P_1}{P_2}$$

$$\boxed{\Delta S = nR \ln \frac{P_1}{P_2}}$$